

```
%%bash
```

```
# --- Configuration ---
```

```
FAT32_PARTITION="/dev/sdb1" # <--- !!! CHANGE THIS TO YOUR ACTUAL FAT32  
PARTITION !!!
```

```
MOUNT_POINT_FAT32="/mnt/fat32_lower"
```

```
OVERLAY_BASE="/mnt/overlay_fat32"
```

```
LOWER_DIR_LINK="${OVERLAY_BASE}/lower"
```

```
UPPER_DIR="${OVERLAY_BASE}/upper"
```

```
WORK_DIR="${OVERLAY_BASE}/work"
```

```
MERGED_DIR="${OVERLAY_BASE}/merged"
```

```
# --- 1. Create necessary directories for OverlayFS ---
```

```
mkdir -p "${MOUNT_POINT_FAT32}"
```

```
mkdir -p "${UPPER_DIR}"
```

```
mkdir -p "${WORK_DIR}"
```

```
mkdir -p "${MERGED_DIR}"
```

```
# --- 2. Mount the FAT32 partition (as lowerdir) ---
```

```
# Using 'vfat' for FAT32, 'nofail' to prevent boot issues if not found
```

```
# 'umask=000' makes files readable/writable by everyone by default - adjust as needed
```

```
# 'iocharset=utf8' for proper character encoding
```

```
# 'shortname=winnt' for Windows NT compatible short filenames
```

```
echo "Mounting FAT32 partition ${FAT32_PARTITION} to ${MOUNT_POINT_FAT32}"
```

```
sudo mount -t vfat -o rw,nofail,umask=000,iocharset=utf8,shortname=winnt
```

```
"${FAT32_PARTITION}" "${MOUNT_POINT_FAT32}"
```

```
if [ $? -ne 0 ]; then
```

```
    echo "Error: Failed to mount FAT32 partition. Please check the device name and ensure it's a  
valid FAT32 filesystem."
```

```
    exit 1
```

```
fi
```

```
# Create a symbolic link for the lowerdir within the OverlayFS structure
```

```
# This ensures the fuse-overlayfs command uses the mounted FAT32 partition as its lowerdir
```

```
ln -sf "${MOUNT_POINT_FAT32}" "${LOWER_DIR_LINK}"
```

```
echo "Contents of mounted FAT32 partition (${MOUNT_POINT_FAT32}):"
```

```
ls -l "${MOUNT_POINT_FAT32}"
```

```
# --- 3. Mount OverlayFS using fuse-overlayfs ---
```

```
# Unmount previous OverlayFS mount if it exists to avoid conflicts
```

```
sudo umount "${MERGED_DIR}" 2>/dev/null || true
```

```
echo "Mounting OverlayFS with FAT32 as lowerdir..."
sudo fuse-overlayfs -o
lowerdir="${LOWER_DIR_LINK}",upperdir="${UPPER_DIR}",workdir="${WORK_DIR}"
"${MERGED_DIR}"
if [ $? -ne 0 ]; then
    echo "Error: Failed to mount fuse-overlayfs. Check permissions and directory setup."
    exit 1
fi

echo "OverlayFS with FAT32 lowerdir mounted successfully at ${MERGED_DIR}"

# --- 4. Verify the mount and contents ---
echo "--- Verifying OverlayFS mount ---"
findmnt -t fuse-overlayfs | grep "${MERGED_DIR}"

echo "--- Listing contents of ${MERGED_DIR} (should show FAT32 files + upperdir files) ---"
ls -l "${MERGED_DIR}"

echo "--- Test: Create a file in the merged view (should go to upperdir) ---"
echo "Hello from OverlayFS" > "${MERGED_DIR}/test_overlay_file.txt"
ls -l "${MERGED_DIR}/test_overlay_file.txt"
ls -l "${UPPER_DIR}/test_overlay_file.txt"

echo "Cleanup (optional): To unmount, run:"
echo "sudo umount ${MERGED_DIR}"
echo "sudo umount ${MOUNT_POINT_FAT32}"
```